Lương Thế Hải (Hai Luong)

Software Engineer | C/C++ · Embedded Systems · AI/ML
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PROFESSIONAL SUMMARY

Results-driven Software Engineer with 4+ years of experience building high-performance systems across embedded devices, game engines, and AI-powered applications. Proficient in C/C++, Python, and modern build systems. Hands-on experience with AI agent development using LangChain, RAG, and Ollama. Strong track record of delivering cross-platform solutions, migrating legacy systems, and driving knowledge-sharing within engineering teams.

TECHNICAL SKILLS

Languages	C, C++, Python, JavaScript, C#
Frameworks & Tools	Qt/QML, LangChain, MFC (libcurl), VueJS, WebAssembly, Docker
AI / ML	LangChain, RAG, Ollama, Embedded Text, OpenCV, TensorFlow, PyTorch
Build Systems	CMake, Makefile, Premake5, GitHub Actions
Methodologies	Spec-Driven Development (SDD), Vibe-Coding, Agile, CI/CD
APIs & Platforms	PythonParts API (Allplan), SDS2, GPIO, REST APIs

WORK EXPERIENCE

Software Engineer | GSI Software *Jun 2026 – Present*

- Conducted R&D on AI Agent solutions for construction engineers, leveraging LangChain, Ollama, RAG, and embedded text retrieval to explore intelligent automation for domain-specific workflows.
- Maintained Allplan BIM plugins using Python and PythonParts API, improving plugin stability and feature coverage.
- Drove internal knowledge-sharing sessions on Data Structures & Algorithms, Design Patterns, and Spec-Driven Development (SDD) methodologies.
- Delivered Vibe-Coding projects leveraging GitHub Spec-Kits and SDD, reducing initial development cycle time.
- Maintained SDS2 structural design project with C++ and Makefile, resolving critical cross-platform build issues.

Software Engineer | FPT Software *Feb 2024 – Jun 2025*

- Led GUI migration of embedded device interfaces from a legacy framework to Qt/QML 6 (C++, CMake), improving rendering performance and maintainability.
- Built and maintained internal Python tooling that streamlined coding and documentation workflows across the engineering team.
- Maintained a GUI REST API client component built with libcurl and MFC, ensuring compatibility with evolving backend APIs.
- Participated in company hackathon, implementing a RAG-based AI solution to accelerate internal knowledge retrieval.

Software Engineer | Glis Games *Feb 2022 – Jan 2024*

- Engineered communication modules between embedded hardware devices and games using C++, GPIO, and manufacturer APIs.
- Developed and maintained cross-platform games with C++, WebAssembly, and JavaScript, targeting both desktop and web environments.
- Contributed to front-end web game development using VueJS, shipping multiple interactive game features.
- Maintained and extended the company's in-house 2D game engine, targeting Windows, Linux, and Web platforms.

- Conducted R&D on single-player game server architecture using C++, C#, and Docker.

Internship | A.N LAB *Dec 2020 – Aug 2021*

- Trained in computer vision and image processing using OpenCV, TensorFlow, and PyTorch, applying models to solve real-world classification and detection tasks.

EDUCATION

Bachelor of Software Engineering *Oct 2018 – Mar 2023*

FPT University | Hanoi, Vietnam

CERTIFICATIONS

Authentication and Authorization for Web/API (LFEL1004) – The Linux Foundation *Dec 2025*

Professional Certificate – CEFR B2 (English) – EnglishScore *Jul 2024*

LANGUAGES

Vietnamese (Native) | **English** (Upper Intermediate – CEFR B2) | **Japanese** (Elementary) | **Chinese (Mandarin)** (Elementary)

NOTABLE PROJECTS

SekaiEngine

- Personal open-source 2D game engine built in C++ from scratch, running on Windows, Linux, and Web (WebAssembly). Showcases low-level engine architecture, rendering pipeline design, and multiplatform CI/CD with GitHub Actions.

Multiplatform C++ CI/CD (Dev.to article)

- Authored a technical article on building and testing cross-platform C++ projects with GitHub Actions, covering matrix builds across multiple OS environments.